Sentiment Analysis-Based Study on Chinese Audiences Acceptance of "*Friends*"

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Github Repository

Introduction

Although *Friends* is very popular in the world, Chinese audiences has significant difference on understanding humorous. We hope to find that, the influences of the accessible and understanding on Chinese audiences. So, based on score and comments, we do this small sample research.

Because the data source is from Douban comment and season rating, and the open IMDb season rating in Kaggle, we analysis through the difference of rating, emotional, topic, and keywords for searching the influence and accessible. Our data analysis pipeline is based on Google Colab T4, and using Spark to connect the data warehouse. Split all project into five step, which are Raw data storage, useful data extract, sentiment analysis, topic classification, and visualization. And now, let us introduce the system design on this project.

System Design

Our system integrates data storage, data processing, machine learning, and visualization tools. The goal of the system is to generate valuable analytical results in an efficient and intuitive manner by structuring and visualizing the collected dataset.

The system begins with data storage. We collected more than 2,000 user comments and stored them in

Azure Blob Storage.We selected Azure Blob Storage because it provided a reliable and scalable storage support for our dataset.

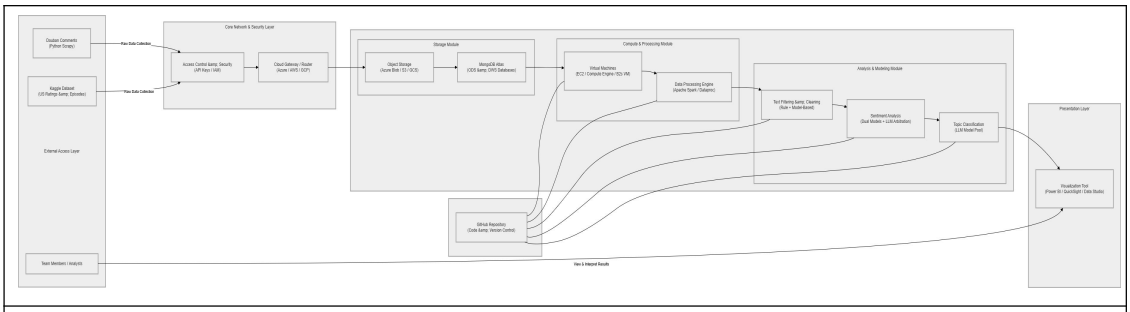
Afterwards, the collected data was transferred to MongoDB Atlas, which served as our main database system. MongoDB Atlas organized and managed the datasetand transformed it into a structured format for later processing and analysis.

The subsequent data processing stage was implemented using Apache Spark. Spark was selected because it supports large-scale data processing, meeting our efficiency requirements for data cleaning and filtering tasks. In this stage, Python was chosen as the primary programming language, and the processing tasks were executed on the Google Colab cloud platform. Python scripts were then used to perform data cleaning and deduplication, preparing the dataset for subsequent machine learning and analysis.

After the data processing stage was completed, the system applied machine learning models such as Logistic Regression and Naive Bayes for analytical tasks. These models were used to classify user comments into different sentiment categories. The use of multiple models further improved the reliability and accuracy of the classification results.

Finally, the processed results were exported to Power BI for visualization. Power BI served as the visualization tool, generating four types of charts, including line charts, bar charts, pie charts, and word clouds. These visualizations provided a more intuitive and accessible way to summarize trends and patterns in audience opinions reflected in the collected user comments.

Overall, our system represents an efficient and scalable workflow that performs data storage, processing, analysis, and visualization within a cloud-based environment.



System Architecture

Budget Plan

To evaluate the feasibility of the proposed system, three vendors were considered for development. The estimated costs included computing reources, database services, data processing, stroage, and visualization tools. All configurations are designed to support he processing of approximately 2,000-2,500 user comments.

Vendor A – Microsoft Azure

Microsoft Azure is selected as the primary cloud platform due to its strong integration with data storage and visualization tools.

Component	Service	Monthly Cost(HKD)
Virtual Machine	Azure B2s VM(2vCPU,4GB RAM)	800
Database	MongoDB Atlas	600
Data Processing	Azure Spark	800
Storage	Azure Blob Storage	200
Visualization	Power BI	100

Total Monthly Cost:HKD2,700

Total Yearly Cost:HKD32,400

The total cost is well below the HKD 200,000 budget limit and is sufficient to process approximately 2,285 user comments.

Vendor B – Amazon AWS

Component	Service	Monthly Cost(HKD)
Virtual Machine	Amazon EC2 t3.medium	900
Database	AWS	600
Data Processing	Spark	900

Storage	Amazon S3	250
Visualization	AWS QucikSight	150

Total Monthly Cost:HKD2,800

Total Yearly Cost:HKD33,600

The AWS-based system provides similar computing capability and supports large-scale data processing using Spark. The total yearly cost remains significantly below the HKD 200,000 budget limit.

Vendor C – Google Cloud

Component	Service	Monthly Cost(HKD)
Virtual Machine	Google Compute Engine	850
Database	MongoDB Atlas	600
Data Processing	Google Dataproc	850
Storage	Google Cloud Storage	200
Visualization	Google Data Studio	120

Total Monthly Cost:HKD2,620

Total Yearly Cost:HKD31,440

The GCP-based configuration offers a cost-effective solution with strong integration for cloud-based data processing and visualization tasks.

Method

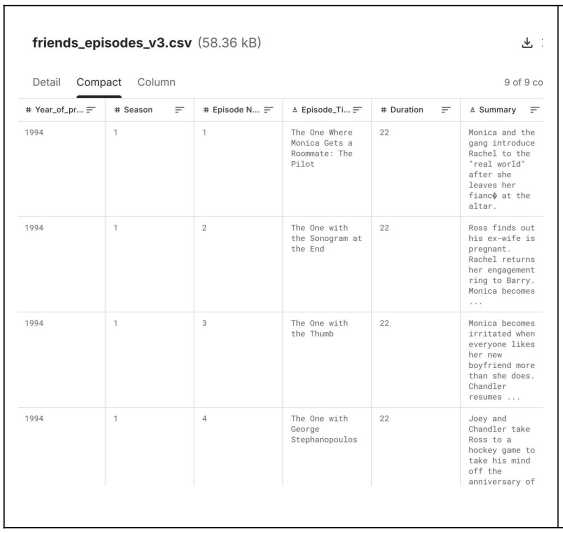
We decided the whole into 4 steps, which are Data collect, Data Extract, Modeling, and Data Visualization.

1. Data Collect (ODS->DWS)

1.1. Data Source

We use douban platform as our data source. Douban as most popular and television review community

in China. We through this platform to scrapy chinese users short comments and rating. And then, in order to compare with the rating between China and America, we get some csv files from Kaggle.

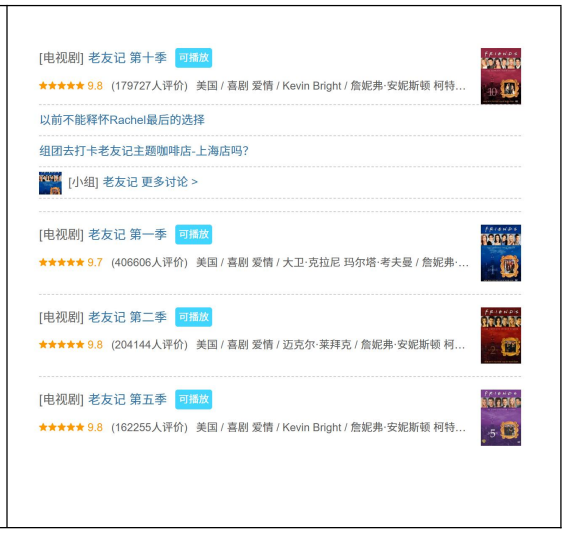


friends_episodes_v3.csv (58.36 kB)

Detail Compact Column 9 of 9 co

Year_of_pr...	Season	Episode N...	Episode_Ti...	Duration	Summary
1994	1	1	The One Where Monica Gets a Roommate: The Pilot	22	Monica and the gang introduce Rachel to the "real world" after she leaves her fianc‹ at the altar.
1994	1	2	The One with the Sonogram at the End	22	Ross finds out his ex-wife is pregnant. Rachel returns her engagement ring to Barry. Monica becomes ...
1994	1	3	The One with the Thumb	22	Monica becomes irritated when everyone likes her new boyfriend more than she does. Chandler resumes ...
1994	1	4	The One with George Stephanopoulos	22	Joey and Chandler take Ross to a hockey game to take his mind off the anniversary of

Kaggle data file



[电视剧] 老友记 第十季 可播放
★★★★★ 9.8 (179727人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...
以前不能释怀Rachel最后的选择
组团去打卡老友记主题咖啡店-上海店吗?
[小组] 老友记 更多讨论 >
[电视剧] 老友记 第一季 可播放
★★★★★ 9.7 (406606人评价) 美国 / 喜剧 爱情 / 大卫·克拉尼 玛尔塔·考夫曼 / 詹妮弗·...
[电视剧] 老友记 第二季 可播放
★★★★★ 9.8 (204144人评价) 美国 / 喜剧 爱情 / 迈克尔·莱拜克 / 詹妮弗·安妮斯顿 柯...
[电视剧] 老友记 第五季 可播放
★★★★★ 9.8 (162255人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...

Douban platform

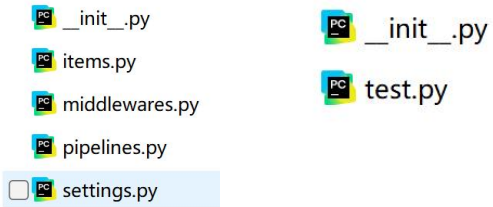
1.2. Acquisition Technology

We use Python scrapy to collect data from Douban platform. We used Python Scrapy as our crawling framework. The basic process of Python Scrapy have 4 steps.

Operation code	Examples of pictures

scrapy startproject+"the name of scrapy file"	`C:\Users\任钱\Desktop>scrapy startproject douban`
	`New Scrapy project 'douban', using template directory`
	`ckages\scrapy\templates\project', created in:`
	`C:\Users\任钱\Desktop\douban`
	
	`You can start your first spider with:`
	`cd douban`
	`scrapy genspider example example.com`

scrapy genspider+"scrapy name"+"crawler website"	`C:\Users\任钱\Desktop\douban>scrapy genspider test www.douban.com`
	`Created spider 'test' using template 'basic' in module:`
	`douban.spiders.test`

Python file edit	
scrapy crawl+"scrapy name"	`C:\Users\任钱\Desktop\douban>scrapy crawl test -o data.csv`

There also have more some detail about Python Scrapy.

Regarding 'Python file edit', the different Python file have different function. There we only talk some basic files.

1) item.py: This file is used to set data schema. It defined data schema and data type so that provide uniform structural standard for data collect, data cleaning and storage.

2) settings.py: This file is used to set project configuration.

3) test.py: This is core crawler implementation file. You need to edit code to achieve crawl data from website to the specified file.

And then talk about scrapy output command.

scrapy crawl+"scrapy name": Look at picture, the '-o data.csv' represent that file name and the type of file. If you want to output json file, you can 'scrapy crawl test -o filename.json'.

1.3.Data Schema & Storage

1) Data Schema

About the data of douban platform, we design 2 tables, a 'ods_douban_comment table' and 'ods_douban_season_rating'. So they have different data schema.

Table 1: ods-douban-comment Fields: id season (int) comment (String) name (String)	Table 2: ods _ douban _ season _ rating Fields: season (string) rating (double) total _ people (int)

ods_douban _comment table ods_douban_season_rating table

Regarding the data that come from Kaggle, since they are already well organized and public accessible, so we preserved the orginal data structure and import data into a table that called 'ods_friends_episodes_v3'.

Table3: ods _ friends _ episodes _ v3
Fields: Year-of-prod (int)
Season (int)
Episode Number (int)
Episode _ Title (String)
Summary (String)
Director (String)
Rating (double)
Votes (int)

ods_friends_episodes_v3 table

2) Data Storage

We stored the collected data in the MongoDB-based ODS database. The full name of ODS is 'Operational Data Store'. This layer is used for temporary storage layer and construct a connection between raw data sources and data warehouse.

1.4.Data Size

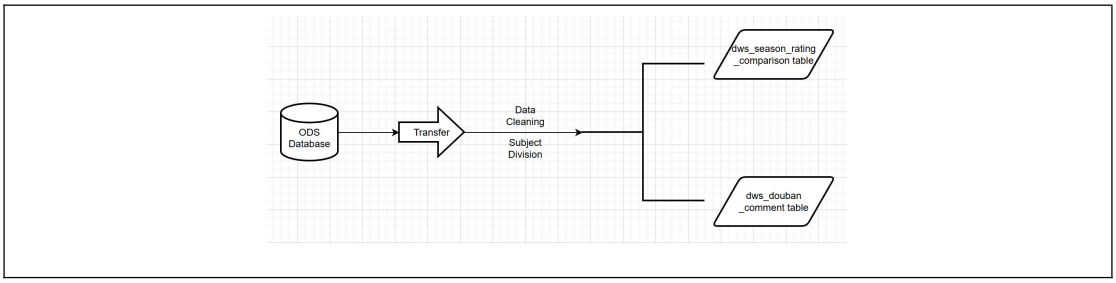
We use Python Scrapy to scrapy 4000 short comments from douban ,then output as CSV file. And we get 10 rating data of chinese audiences. Also we get 10 rating data of American audiences.

2. Data Extract

2.1. Process

Focus on Data Extract, we followed the path that from ODS to ETL process, and eventually to the DWS layer.

The full name of ETL is 'Extract, Transform, Laod'. The process that combines data from mutiple data sources into a single. In our project, we extracted some important data and loaded into MongoDB-based DWS database. We built this database through topic aggregation and data cleaning based on the ODS layer data.



The workflow of Data Extract

2.2. Detail

1) Data cleaning

In this project, during the ETL process from the ODS to the DWS layer, we first perform data deduplication, followed by sorting the records in ascending order based on the TV series season.

2) Subject divison

Based on the field topic of ODS layer, we divide and summarize the data into two core topic tables: rating comparison table and comment table. The first table aggregates rating indicators of each season to quantify audience feedback, and the second table aggregates audience comments of each season. After data cleaning and unifying the indicator caliber, we finally form a standardized and efficient indicator query service in DWS layer, providing solid data support for modeling and analyzing the acceptance of *friends* in China in ads layer.

3) Some code

In order to improve the code readability and data search effeciency, we use some sql statement.

```
#生成 dws_season_rating_comparison表
df_dws_rating=duckdb.query("""
  SELECT d.season,
      d.rating AS df_douban_rating,
      i.imdb_rating,
      ROUND((d.rating-i.imdb_rating),1) AS rating_difference
  FROM df_ods_douban_rating d
  LEFT JOIN (SELECT ROUND(AVG(Stars),1) AS imdb_rating,
             Season
         FROM df_ods_imdb_rating
         GROUP BY Season) i
  ON d.season=i.Season
  ORDER BY d.season
""").df()
```

```
#生成 dws_douban_comment
df_dws_comment=duckdb.query("""
  SELECT id AS user_id,
      season,
      "name" AS user_name,
      comment
  FROM df_ods_comment
  ORDER BY season
""").df()
```

2.3. Subject Table Construction

We constructed 2 subject table: a 'dws_season_rating_comparison' table and a 'dws_ douban_comment' table. They have different function.

1) dws_season_rating_comparison: Aggregates rating indicators of each season to quantify audience feedback.

2) dws_ douban_comment: Aggregates audience comments of each season.

Table 1. dws _ season _ rating _ comparison Fields: season (int) douban-rating (double) imdb-rating (double) rating -difference (double)	Table 2: dws _ douban _ comment Fields: user _ id name (String) comment (String) season (int)
dws_season_rating_comparison table	dws_douban_comment table

3. Modeling and Analysis Pipeline

In out project, the modeling and analysis pipeline is called ADS layer, it's detail name is application data services. It's the core analyze module, and the main function of it is get the Pre-cleaning data from DWS layer, and then turn the data into structure data with analysis result. These result will used to interpretation and visualization directly.

This pipeline contains seven steps, which they are: Rule-based Filtering, Model-based Filtering, Pre-processing Text, Sentiment analysis, topic classification, Integration of Sentiment and Topic Results, and then output the final tables. The detailed information explanation follows.

3.1 Rule-based Filtering

First stage is rule-based filtering. The main goal is to remove low quality comments before they go to models. We do things like: delete NaN, fix text format, and remove useless comments (like only symbols or numbers). Also, we delete duplicate records here. This is a basic data quality control. It uses the lowest compute to find useful data, which helps us save time and money for the later model processing.

3.2 Model-based Filtering

After the rules, we still need model-based filtering. This part is to find spam, offensive words, or comments with no real meaning. We don’t use single classifier; we use a multi-model integrated way to make the result more stable and reliable. After this step, the comments are much better for sentiment

analysis and topic classification. This makes our final result more trustworthy.

3.3 Text Pre-Processing

The third step is the text pre-processing. In this step, filtered comments will then enter the text preprocessing stage to further improve the consistency and analysis ability of the text. This stage mainly includes Chinese word segmentation, stop word removal, and basic text normalization. Withing this step, the comment text will be transformed into a form more suitable for model processing, providing higher-quality input data for subsequent sentiment analysis, keyword extraction, and topic classification.

3.4 Sentiment Analysis

This step involves sentiment analysis of all comments. Considering that short Chinese comments are often brief and semantically incomplete, a single model is prone to instability or bias in sentiment assessment. Then, we employ a multi-model approach for sentiment evaluation. By combining the results of different models, the stability and reliability of the analysis can be improved.

For comments with inconsistent model judgments or unclear sentiment tendencies, we further introduce an additional judgment mechanism for arbitration to reduce errors from a single judgment. This allows for more effective handling of comments with complex tones, ambiguous expressions, or mixed emotions.

After all, each comment is uniformly categorized into three sentiment labels: positive, neutral, and negative.

3.5 Topic Classification

Beyond assessing the sentiment of the comments, we further categorizes the content into themes. Based on the project's analysis objectives, the comments are primarily divided into three core themes: acting, plot, and nostalgia. Through this step, we can not only determine whether viewers' attitudes towards

Friends are positive or negative, but also further identify the key points of discussion, thus providing a more comprehensive explanation of the sources of audience acceptance.

3.6 Integration of Sentiment and Topic Results

After completing sentiment analysis and topic classification, we integrate the two results into a unified comment-level analysis table, and it contain all fields. But before merging, the topic classification results are deduplicated to ensure that each comment retains only one valid topic record. Because data duplication was possible under the previous high concurrency conditions. Then, we merged the sentiments analysis results and topic classification results. Now, every comment has both "sentiment" and "topic" tags. After this, the data structure is more complete. It is much easier for us to do the following statistical analysis and visualization.

3.7 Output Tables

At the end, we have two core result tables for the next steps.

The first one is the comment-level table. This table keeps everything for each comment: season, text, keywords, sentiment tags, and topic category. It helps us look into the details of every single comment and see how keywords and emotions work together.

The second one is the season-level summary table. It counts the sentiment distribution for each season and compares it with the platform ratings. This table is better than the first one if you want to see the overall trend—like how Chinese audiences' feelings about *Friends* changed over time.

These two tables are the data foundation for our Power BI charts and final report. They turn the ADS layer data into real findings that everyone can understand.

Visualization

To gain a more intuitive understanding of audiences' reactions and discussion topics related to Friends, we conducted sentiment analysis and topic classification on the data, and then presented the final results through data visualization.We used Power BI to process the dataset and ultimately generated four visualization charts. The visualization process mainly focused on audience rating comparisons for

Friends, discussion topic directions, and sentiment tendency analysis. Keywords were aggregated and summarized, transforming the dataset into more intuitive visual representations.

The first visualization chart is a line chart that compares audience ratings from two platforms. The line chart clearly shows that ratings on Douban are generally higher than those on IMDb. This result reflects differences in audience perceptions of Friends between Chinese and American viewers, with Chinese audiences tending to give higher overall ratings.

The second visualization chart is a bar chart that summarizes various discussion topic keywords found in user reviews. The results show that comments related to the plot appear most frequently, followed by nostalgia and then actors. This finding indicates that audiences pay the greatest attention to plot development compared to other aspects, while actors' performances and emotional resonance are also important topics of discussion.

The third visualization chart is a pie chart that illustrates the distribution of sentiment in audience reviews. The chart shows that the majority of comments express positive sentiment, a smaller portion reflects neutral sentiment, and only a very small proportion represents negative sentiment. This result indicates that audiences generally hold a positive and supportive attitude toward Friends.

The final visualization chart is a word cloud generated from high-frequency keywords appearing in audience reviews. Words such as "love," "nostalgia," "friends," and "touching" appear most frequently. This result summarizes the key elements that audiences value most in the series.

Overall, the visualizations presented intuitive support for the analysis in this project and strongly demonstrate that Friends maintains a high level of acceptance among Chinese audiences.

Result And Discussion

The results of rating comparison, sentiment analysis, topic classification, and keyword analysis suggest that *Friends* has a relatively high level of acceptance among Chinese audiences. In ratings, Douban scores are generally higher than those on comparable platforms such as IMDb. At the same time, the negative comments is truly little, the positive comments are larger even more than negative comments. These data all point to positive attitude to this drama between Chinese audience. And this is still exist in the discussion content. The topics classification shows that, the audience are mainly talk about the

plot, nostalgia, and character. In topics, the plot has the highest discussion rate, reflect that the stories is more important within audience. No only that, the keywords analysis also confirm about the point. Just like "classic", "like" and the name of characters is high-frequency words. In this way, it an be explained from the front that *Friends* can be regarded as a good drama. In summary, whatever in comments, or rating, they all support a conclusion: Chinese audience like *Friends* not only reflected at high rate, but also on positive emotional and plot.

Conclusion

Overall, this study shows that *Friends* among Chinese audiences has high over acceptance and provides an overall evaluation in Chinese audiences.

At the same time, this project built a relatively complete analysis pipeline. From data collection, data extraction, to ADS text analysis, ending with Visualization, showing a workflow from raw data to complete output.

However, this project still exists some problem, such as the data source is too concentrated, the result of text classification could be affected by the complexity of comments. In the future, we can expand the data size, collect data from different platform, and refine the analysis methods to improve the representativeness and stability.

Group Contribution

Hu Jihan: Make model for text analysis
Chen Jiahao: Data Collect and Data Extraction
Tang Chao: Data Visualization and System

GitHub Repository

Project GitHub Repository [*https://github.com/mgHurryo/STAT_2630SEF_Group_Project*]

References

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4. GitHub, for open-source NLP libraries including jieba, KeyBERT, and SnowNLP; and stopwords-zh liabrary.
5. DeepSeek API, for LLM-based arbitration in sentiment analysis.
6. AliYun API, for LLM-based topic classification and model pool support.
7. Moonshot API, for supplementary LLM-based topic classification.